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Section: C
Assignment: 01

Question:

You are given the responsibility to design a small network (LAN) of 10 computers that is supposed to be used among all of your friends to share lecture materials and necessary academic files.

As you have learnt about various kinds of physical topologies, which of the topologies will be the best for your design of this network? Show the design drawing a diagram and label different components in it.

Justify your answer with sound rationale (why you have decided to use that topology).

Answer:

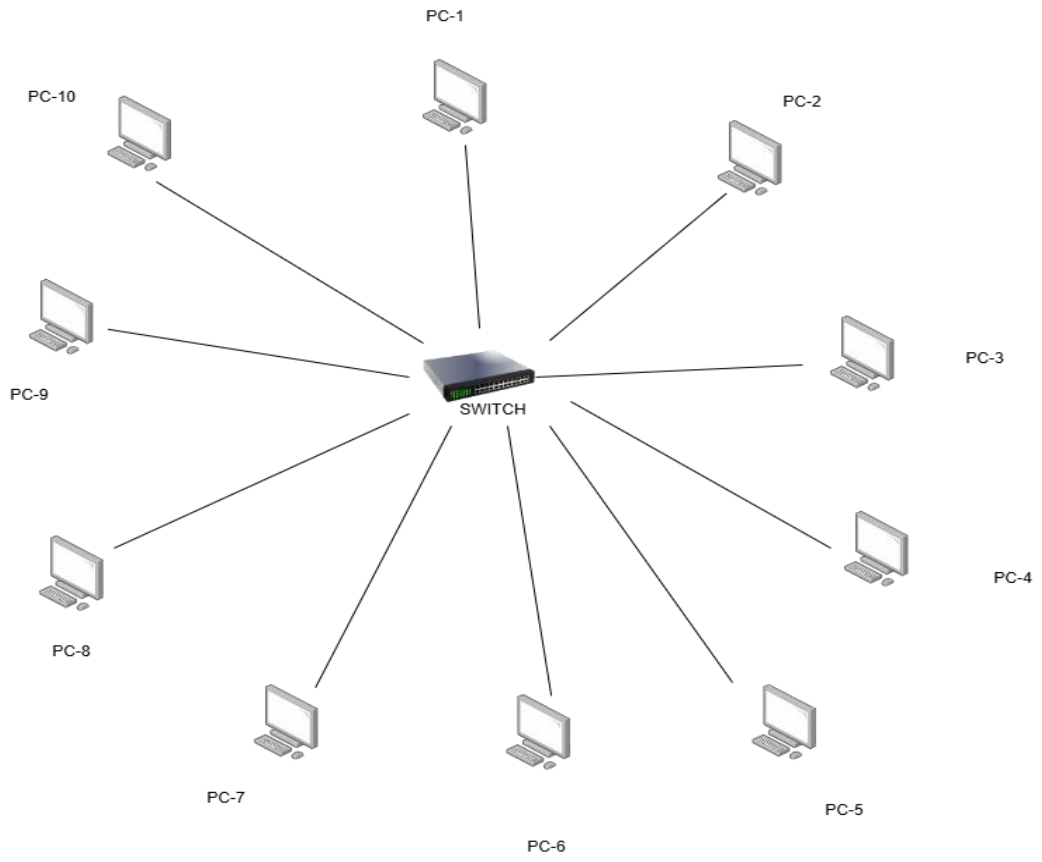


Figure: Star Topology LAN Design for 10 Computers

A Local Area Network is used to connect multiple computers to share data and resources efficiently. For a network of 10 computers, the most suitable topology is the star topology.

In this design, all computers are connected to a central device called a network switch using Ethernet cables. Each computer is labeled from PC 1 to PC 10. The switch controls communication and sends data only to the intended device. Star topology is chosen because it is easy to manage and reliable. If one computer fails, the rest of the network continues to work normally. It also provides better performance and reduces network traffic. Another advantage is that new computers can be added easily, and troubleshooting is simple due to separate connections.

In conclusion, star topology is the best choice for this LAN because of its simplicity, reliability, and efficiency.